

Discussion Module:

DSE Analytics, GraphX,
GraphFrames

Discussion/Information Only-

- Covered in detail when we cover DSE Graph

DSE Analytics: GraphX, GraphFrames

Allows Spark to natively manage graphs

- In general we don't address the GraphX APIs, but instead integrate with GraphFrames; GraphFrames is not GraphX
- GraphX is to RDDs as GraphFrames are to DataFrames
- GraphX extends the Spark RDD, GraphFrames extends Spark's DataFrames
- Developed by DataBricks in conjunction with UC Berkeley and MIT
- GraphFrames fully integrate with GraphX via conversions between the two representations, without any data loss
- If a GraphX API is more useful, you are free to convert back and forth
- With DSE Graph integration with GraphFrames, seamless integration of DSE Graph with Spark
- Allows for certain type of graph jobs to run more efficiently in Spark
- Spark GraphFrames API uses familiar terms to the Tinkerpop API making transition easier

DSE Analytics: GraphX, GraphFrames

DseGraphFrames:

- DSE Graph data using native Spark mechanisms
- Execute bulk type processing on DSE Graph data
- Combine DSE Graph and non Graph data
- Access DSE Graph through SQL like APIs
- Analytic jobs on graphs

Mimics TinkerPop3 traversal language

- API's in; Scala, Java, (some Python)
- Implicit conversions from GraphFrame to DseGraphFrame and back
- GraphFrame filtering and TinkerPop methods could be mixed

DSE Analytics: GraphX, GraphFrames, Exercise

- In Studio load the graph studio worksheet from, <https://github.com/riptano/enablenment/tree/master/boot-camp-dse/studio>
- Graph name can be whatever
- Create the schema
- Following the next line of instructions from the worksheet using the spark shell
- Download data
- Import data through spark using GraphFrames
- Simple queries
- Once in studio
- Try to replicate the query using GraphFrames
- Try to get a sense on how they are different
- Export the graph in DSEFS using parquet



End of Module: